CoAP Non-traditional Responses

2026-07-06, <draft-ietf-core-responses-01.txt>

"Traditional": CoAP [RFC7252]: a single response is sent, on the same transport, to the client that posed a request

- RFC7252, core-groupcomm-bis: responses to multicast requests
- RFC7641 multiple responses following an observation request
- RFC7967 not sending a response at all
- RFC9177 requesting a series of response blocks
- core-groupcomm-proxy, core-observe-multicast-notifications ...

Terms

Non-traditional response: A response that is **not the single response** generated for a request received on the same transport.

Non-matching response: A response that is incompatible with the original request, and thus in particular unsuitable as a cached response to that request (but possibly suitable to populate the cache for a similar request).

Configured request: A request that reaches the server in another way than by transmitting a usual CoAP request on the same communication channel a response is expected on.

Embedded request: A request that is provided by the server to the recipient of its response by **embedding it into the response**.

CoAP Options Proposed in Draft

Response Option:

- **`Response-For`**, for **embedding** requests in a response
  - uses an OSCORE-like form

Request Options:

- **`Leisure-For-Responses`** to indicate **client is prepared** for additional responses
- **`Respond-To`** for the requesting a non-matching response (to someone else)
  - define safe cases: security implications depend on context

What Can We Move Forward?

Can be used today:

- **`Leisure-For-Responses`**, for a client to enable additional responses, and
- **`Response-For`**, for embedding a request in a response

Further work is needed on:

- **`Respond-To`** for soliciting responses for configured requests
  ➔ discussed in Appendix B.

Questions to the WG

- Are we ready to go ahead with embedded requests as in **`Response-For, Leisure-For-Responses`**
  - even if the latter might occasionally need to be specialized?
- Are we happy to keep **`Respond-To`** on the back burner?
  - Is it OK to have a discussion appendix
    or should we cut that out until it's ready?